MILITARY SPECIFICATION

LEADS, ELECTRICAL (STORAGE BATTERY)

This specification has been approved by the Department of Defense for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers electrical leads, for storage batteries. Leads are round, insulated, single conductor, stud mounted for both positive and negative application. (See 6.1.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification.

SPECIFICATIONS

FEDERAL

WW-T-799 — Tubing, Copper, Seamless (for use with solder joint or flared tube fittings)

PPP-B-585 — Boxes, Wood, Wirebound.

PPP-B-591 — Boxes, Fiberboard, Wood Cleated

PPP-B-601 — Boxes, Wood Cleated, Plywood

PPP-B-621 — Boxes, Wood, Nailed and Lock Corner

PPP-B-636 — Boxes, Fiber

PPP-T-76 — Tape, Paper, Water Resistant, Pressure-Sensitive

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MIL-B-116 — Preservation, Methods of

MIL-B-138 — Boxes, Wood, Fiber-Lined for Overseas Shipment

MIL-C-13486 — Cable, Vehicle; Low Tension, 30 Volt Heavy Duty. Single Conductor and Multiconductor

STANDARDS

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 — Carking for Shipment and Storage

MS-35915 — Lead, Electrical (Storage Battery)

(Copies of specifications and standards required by contractors in connection with specific procure-

FSC 6140

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ment functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE

Consolidated Freight Classification Rules

(Applications for copies of the above publications should be addressed to Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Leads. Leads shall consist of a single insulated conductor with terminals attached to each end.

3.1.1 *Conductors.* Conductors for leads shall be in accordance with type I of Specification MIL–C–13486.

3.1.2 *Terminals.*

3.1.2.1 *Material.* Material for terminals shall be copper in accordance with Specification WW–T–799.

3.1.2.2 *Finish.* Finish for terminals shall be lead plating .004 ± 001 inch thick.

3.1.2.3 *Design.* Except as specified in 3.1.2.4 and 3.2 design for terminals shall be optional.

3.1.2.4 *Tubular area.* Terminals shall be provided with an inside tubular area which will permit the assembly with the appropriate wire size.

3.2 Dimensions. Dimensions and tolerances required in the fabrication of leads shall be in accordance with MS 35915 or applicable drawing as specified by the procuring activity (See 6.2). All dimensions shall include the specified plating.

3.3 Mechanical strength. After assembly leads shall show no signs of breakage or distortion when subjected to a pull of 300 pounds.

3.4 Workmanship. Leads shall be free from burrs, sharp edges and other imperfections which may impair serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling. Sampling, including lot size, shall be in accordance with Standard MIL–STD–105. The AQL shall be 1.0 percent for major defects and 4.0 percent for minor defects.

4.2 Acceptance inspection. Acceptance inspection shall be conducted by the Government or under the supervision of the Government.

4.2.1 *Inspection conditions.* All inspection shall be conducted at room ambient temperature, pressure and humidity.

4.2.2 *Classification of defects.* Major and minor defects shall be as defined in table I.

4.3 Test Procedures.

4.3.1 *Visual and mechanical.* Leads shall be inspected for dimensions, finish and workmanship for conformance with the requirement paragraph listed in table I.

4.3.2 *Mechanical strength.* Mechanical strength of leads shall be determined by fully hanging weights or a suitable testing machine. Leads shall be firmly clamped by means of a bolt and a washer. The bolt shall be assembled to the test equipment through the stud hole of the terminals and a direct pull of 300 pounds applied parallel to the longitudinal axis for 1 minute. Leads shall then be inspected for conformance with 3.3.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

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TABLE I

Inspection	Requirement paragraph	Test paragraph	Classification of Defects	
			Major	Minor
Visual and Mechanical	3.1 to 3.2 incl.	4.8.1		
Stud hole dia.	3.2	4.3.1	X	
Dimensions (except stud hole)	3.1.2.4 3.2	4.3.1		X
Finish	3.1.2.2	4.3.1	X	
Mechanical Strength	3.3	4.3.2	X	
Workmanship	3.4	4.3.1		X

5.1.1 *Level A.*

5.1.1.1 *Cleaning.* Cleaning shall be accomplished to remove any deposit foreign to the composition of the part in accordance with process C-1 of Specification MIL-P-116.

5.1.1.2 *Drying.* The leads shall be dried in accordance with any drying procedure specified in Specification MIL-P-116.

5.1.1.3 *Unit packaging.* The leads shall be unit packaged in accordance with method III of Specification MIL-P-116.

5.1.1.4 *Intermediate packaging.* Identical leads (leads of the same stock number) packaged as specified in 5.1.1.3 shall be further packaged in a close-fitting intermediate container conforming to Specification PPP-B-636, class 2, W5s, any style except FTC. Leads measuring up to 18 inches in length shall be packaged in quantities of 20. Leads measuring over 18 inches in length shall be packaged in quantities of 10. Seams, joints and closures of intermediate containers shall be taped with material measuring 2 inches in width conforming to the requirements of Specification PPP-T-76. The gross weight shall not exceed 20 pounds.

5.1.2 *Level C.* Cleaning, drying and packaging shall be in accordance with manufacturer's commercial practice. Protection must be such as to prevent deterioration during shipment and subsequent intervals preliminary to use.

5.2 Packing.

5.2.1 *Level A.* Identical leads (leads of the same stock number) packaged as specified in 5.1.1.4 shall be packed in a wirebound wood, fiberboard wood-cleated, wood cleated-plywood, nailed wood or a fiberboard-lined wood box conforming to the requirements of Specification PPP-B-585, class 3, PPP-B-591, Overseas type, PPP-B-601, Overseas type, PPP-B-621, class 2 — Overseas and MIL-B-138, respectively. Closure shall be as specified in the applicable box specification or appendix, thereto. The gross weight shall not exceed 200 pounds.

5.2.2 *Level B.* Identical leads (leads of the same stock number) packaged as specified in 5.1.1.4 shall be packed in a wirebound wood, fiebrboard wood-cleated, wood cleated-plywood or a nailed wood box conforming to the requirements of Specification PPP-B-585, class 1, PPP-B-591, Domestic type, PPP-B-601, Domestic type and PPP-B-621, class 1 — Domestic, respectively. Closure shall be as specified in the applicable box specification or appendix, thereto. The gross weight shall not exceed 200 pounds.

5.2.3 *Level C.* Leads packaged as specified in 5.1.2 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

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5.3 Marking.

5.3.1 *Normal markings.* Unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of Standard MIL–STD–129.

6. NOTES

6.1 Intended use. Leads are intended for mounting to lug terminals which have been attached to SAE tapered post of the battery.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Title, number and date of military standard or applicable drawing (see 3.2).

(c) Military standard part number.

(d) Level of required packaging and packing.

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